

Remote host
authoritative logical-role inventory

control, status, supervision, and data

Trusted instrument Ethernet network

direct science-data
path to the host

Main board
endpoint

Video board
endpoint

Other function-board
endpoint

Bridge-board
endpoint

Each active board has its own physically accessible UART service interface
for persistent identity, bootstrap-network settings, factory data, and recovery

The UART paths are separate local service connections, not a shared backplane bus.
Main does not proxy ordinary function-board traffic.